

Reflection Journal – Image Processing Lab

ITAI-1378

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This lab helped me understand images in a deeper way than before, and it changed how I think about pictures on screens. Before this class, I thought images were just photos we see, but now I know they are actually numbers arranged in a matrix. The most surprising discovery for me was learning that every image is made of pixels, and each pixel is just a value (or values) that the computer can change using math. Seeing pixel values and matrices made images feel more technical, but also more understandable.

Another important thing I learned is how mathematical operations directly create visual effects. For example, when we change brightness or contrast, we are just adding or multiplying numbers in the pixel matrix. Blurring, sharpening, and edge detection also come from simple math using kernels and filters. It was interesting to see that strong visual effects come from very basic calculations. The most challenging technique for me was convolution with kernels, because it was hard at first to understand how one small matrix moving across an image can change the whole picture. After seeing the results, it became clearer, but it still took time to understand.

The lab also helped me connect traditional image processing to AI tools like Nano Banana. In class, Nano Banana looked very advanced, but now I understand that it uses the same basic ideas, such as filtering, edge detection, and color changes. The difference is that AI learns the best filters automatically instead of us writing them by hand. This connection made AI tools feel less like magic and more like smart systems built on simple rules.

I can imagine many real-world applications for these techniques. They can be used in phone cameras to improve photos, in medical images to see details better, in security systems to detect movement, and in self-driving cars to understand the road. In a future project, I think it would be interesting to combine traditional methods with AI. For example, using basic image processing to clean or prepare images before giving them to an AI model could improve results and make the system faster.

On a personal level, the part of image processing that interests me most is feature detection, like finding edges or objects in an image. This feels like the bridge between simple images and real AI understanding. This lab changed how I see digital photography and image editing. Now I know that every filter or edit is just math behind the scenes. I still

have questions about how deep learning models choose the best filters and how they learn from data, and I hope to explore that more in future classes.